

# A rapid algorithm for inferring latent mixture structure in replicate social science data

Nathan Kellerman

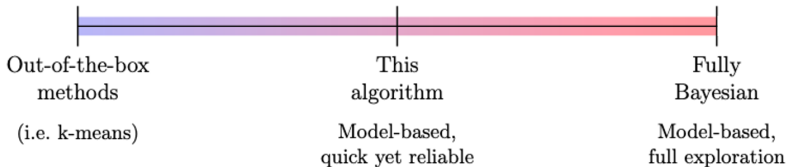
Bowdoin College · Department of Mathematics



Fast, approximate



Slow, expressive



Out-of-the-box  
methods

(i.e. k-means)

This  
algorithm

Model-based,  
quick yet reliable

Fully  
Bayesian

Model-based,  
full exploration

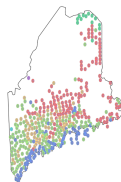
## Theory

- ▶ Professor pedagogy toy example
  - ▶ How can we uncover hidden voting populations in data?
- ▶ Political science framing
  - ▶ Can we mathematically model voting ideology in referendum?

$$\log \mathcal{L}(\vec{\theta}; y)$$

## Application

- ▶ Simulation studies
  - ▶ How rapid and accurate is the algorithm?
- ▶ Case studies
  - ▶ Maine referendum voting
  - ▶ Switzerland referendum voting (previously inaccessible)



**Assume:** Bowdoin College has 200 faculty, each with unique membership to one of 33 academic departments



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3. All students should pursue undergraduate research

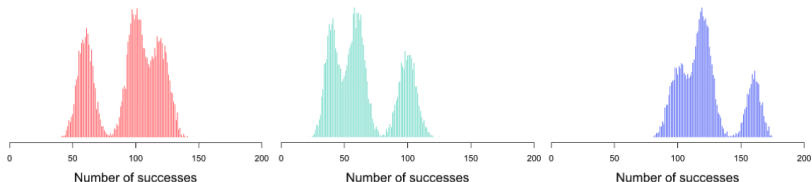
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|               | $Q_1, \text{ Yes}$ | $Q_1, \text{ No}$ | $Q_2, \text{ Yes}$ | $Q_2, \text{ No}$ | $Q_3, \text{ Yes}$ | $Q_3, \text{ No}$ |
|---------------|--------------------|-------------------|--------------------|-------------------|--------------------|-------------------|
| Department 1  | $y_{1,1}$          | $n_{1,1}$         | $y_{1,2}$          | $n_{1,2}$         | $y_{1,3}$          | $n_{1,3}$         |
| Department 2  | $y_{2,1}$          | $n_{2,1}$         | $y_{2,2}$          | $n_{2,2}$         | $y_{2,3}$          | $n_{2,3}$         |
| $\vdots$      | $\vdots$           | $\vdots$          | $\vdots$           | $\vdots$          | $\vdots$           | $\vdots$          |
| Department 33 | $y_{33,1}$         | $n_{33,1}$        | $y_{33,2}$         | $n_{33,2}$        | $y_{33,3}$         | $n_{33,3}$        |

# The shape of professor pedagogy



1. I train my students for a professional workplace
2. Learning should always be difficult
3. All students should pursue undergraduate research

What cultural archetypes of professors exist at Bowdoin?

# Three big questions

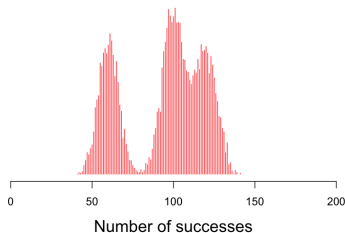
1. How many educational ideologies exist among Bowdoin faculty?
2. How are departments mixed by ideology?
3. What parameters define these ideologies?



- ▶  $i = 1, \dots, M = 33$  departments
- ▶  $q = 1, \dots, Q = 3$  referendum questions
- ▶  $y_{iq}$  yes responses, per department, per question
- ▶  $N_{iq}$  polled faculty per department, per question
- ▶  $k = 1, \dots, K$  latent cultures
- ▶  $z_i \in \{1, \dots, K\}$  latent assignment  $z$  per department



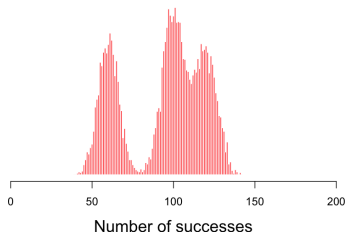
Q1 Support Counts



$$Y \sim \lambda_1 \cdot \pi_1 + \lambda_2 \cdot \pi_2 + \lambda_3 \cdot \pi_3$$

- ▶  $\lambda_1 + \lambda_2 + \lambda_3 = 1$
- ▶ each  $\pi_k$  arises from some well-behaved distributional family  $\mathcal{T}(\theta)$

Q1 Support Counts



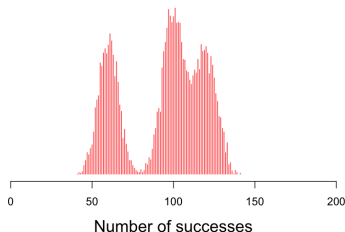
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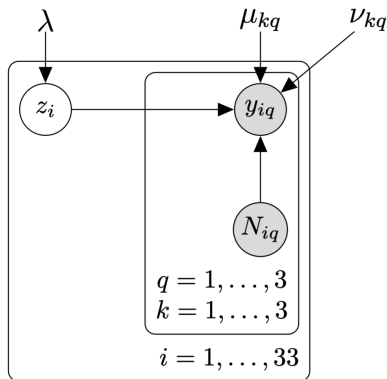
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$$f_k(y_i) = \text{BetaBinomial}(y_i | N_i, \mu_k, \nu_k)$$

Across all  $Q = 3$  questions:

$$\pi(\vec{\theta}_k) = \prod_{q=1}^Q f_{kq}(y_{iq})$$



Data likelihood  $\mathcal{L}(\vec{\theta}, y)$  across  $K$  cultures:

$$\sum_{i=1}^M \log \left\{ \sum_{k=1}^K \lambda_k \left( \prod_{q=1}^Q \text{BetaBinomial}(\mathbf{y}_{iq} | N_{iq}, \mu_{kq}, \nu_{kq}) \right) \right\}$$

### Bayesian approach

- ▶ Examine  $\mathbb{P}(\theta|\mathcal{D})$
- ▶ Find distributions of parameters  $\theta$  by conditioning upon the data
- ▶ Explore all parameters

### Frequentist approach

- ▶ Examine  $\mathbb{P}(\mathcal{D}|\theta)$
- ▶ Construct data likelihood  $\mathcal{L}(\vec{\theta}, y)$
- ▶ Converge on single set of parameters  $\theta$

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| Case Study                  | # of Locales | # of Questions | Polling Year Range |
|-----------------------------|--------------|----------------|--------------------|
| Pedagogy Toy Example        | 33           | 3              | NA                 |
| Maine Referendum Data       | 423          | 70             | 2008 – 2024        |
| Switzerland Referendum Data | 2048         | 390            | 1981 – 2026        |

Rapid algorithm  $\equiv$  maximum likelihood estimation procedure

**Approach:** Try to take derivatives

$$\frac{\partial}{\partial \mu_{kq}} \left[ \sum_{i=1}^M \log \left\{ \sum_{k=1}^K \lambda_k \left( \prod_{q=1}^Q \text{BetaBinomial}(\mathbf{y}_{iq} | N_{iq}, \mu_{kq}, \nu_{kq}) \right) \right\} \right]$$

**Fact:** Finding  $\arg \max_{\vec{\theta}} \log \mathcal{L}(\vec{\theta}; X)$  is hard

Let's “pretend” to know  $\mathcal{Z}$  and let our data guide us in the right direction

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Consider  $\mathcal{D} = (X, \mathcal{Z}) \equiv (\text{observed } X, \text{unobserved latent assignments } \mathcal{Z})$

**Definition:** the complete data log-likelihood is  $\log \mathcal{L}(\vec{\theta}; X, \mathcal{Z})$



**Approach:** Try to take derivatives

$$\frac{\partial}{\partial \mu_{kq}} \left[ \sum_{i=1}^M \log \left\{ \sum_{k=1}^K \lambda_k \left( \prod_{q=1}^Q \text{BetaBinomial}(\mathbf{y}_{iq} | N_{iq}, \mu_{kq}, \nu_{kq}) \right) \right\} \right]$$

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**Claim:** Finding  $\arg \max_{\vec{\theta}} \log \mathcal{L}(\vec{\theta}; X, \mathcal{Z})$  may be easier

**Useful definition:** The probability that department  $i$  is in cluster  $k$  is

$$\gamma_{ik} = \mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1}) = \frac{\lambda_k \prod_{q=1}^Q \text{BB}(x_{iq} | N_{iq}, \mu_{kq}, \nu_{kq})}{\sum_{k=1}^K \lambda_k \prod_{q=1}^Q \text{BB}(x_{iq} | N_{iq}, \mu_{jq}, \nu_{jq})}$$

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**Amazing fact:**  $\mathcal{L}(\theta_t; X) \geq \mathcal{Q}(\theta_t, \theta_{t+1})$ , for

$$\mathcal{Q}(\theta_t, \theta_{t+1}) = \sum_i \sum_k \mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1}) \log \frac{\mathbb{P}(X_{iq} = x_{iq}, Z_i = k | \theta_t)}{\mathbb{P}(Z_i = k | x_{iq}, \theta_{t+1})}$$

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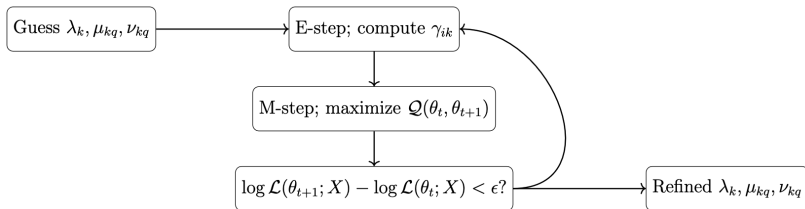
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One EM iteration:

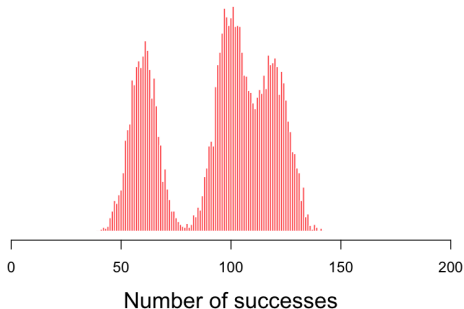
1. E-Step: Compute responsibilities  $\gamma_{ik}$  for each department
2. M-Step: Find the updated parameters which maximize  $\mathcal{Q}(\theta_t, \theta_{t+1})$  and thus improve the likelihood  $\mathcal{L}(\theta; X)$ .

## Flow diagram



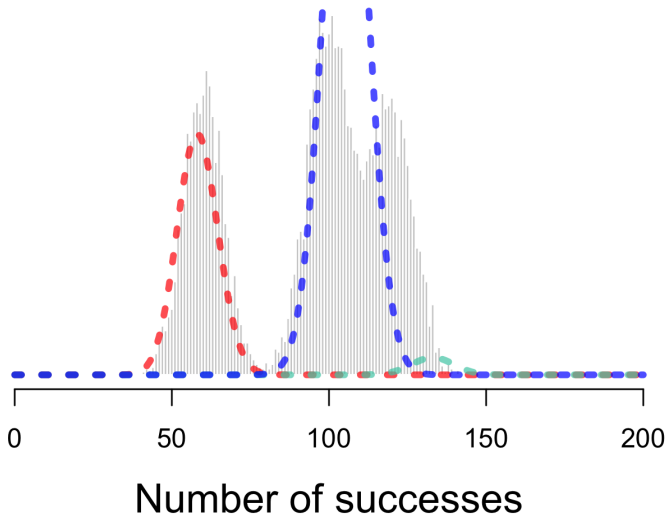
Question 1: I train my students for a professional workplace

### Q1 Support Counts

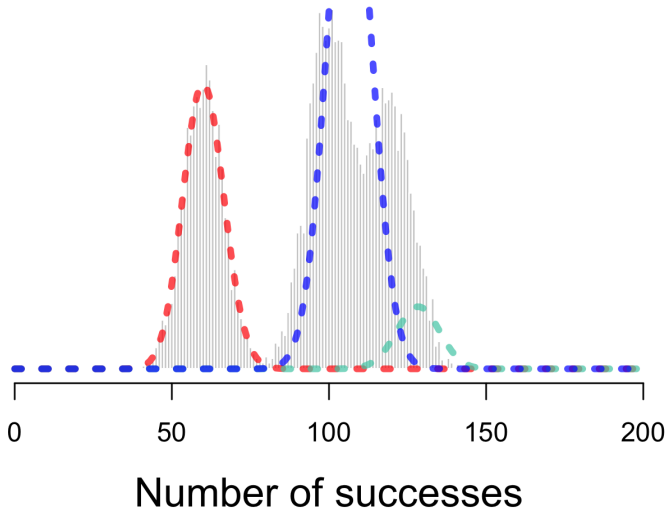


Choose  $K = 3$ . Guess parameters  $\lambda_k$ ,  $\mu_k$ , and  $\nu_k$ .

## EM Iteration 1

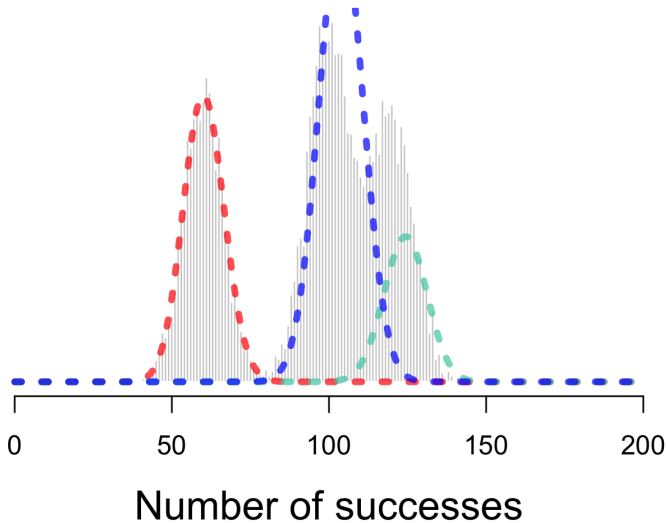


## EM Iteration 2

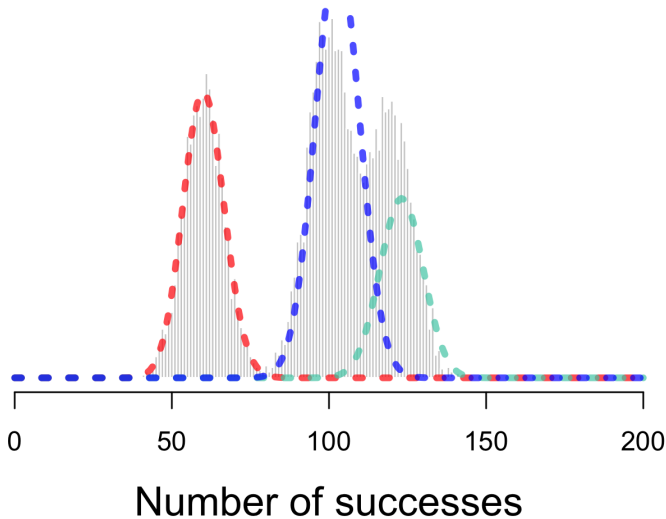




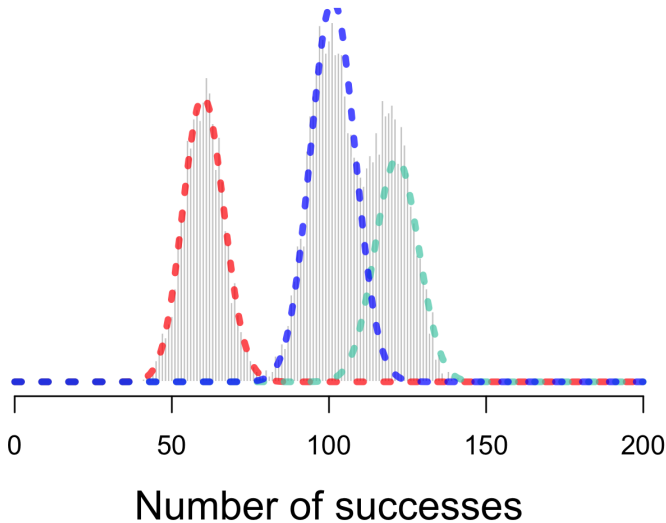
## EM Iteration 4



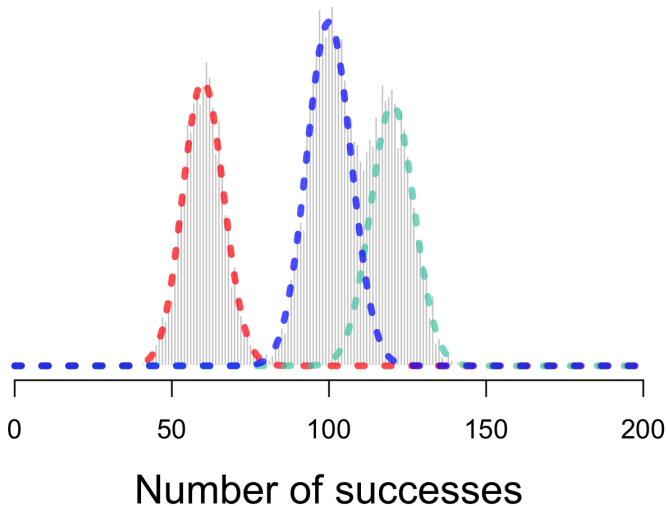
## EM Iteration 5



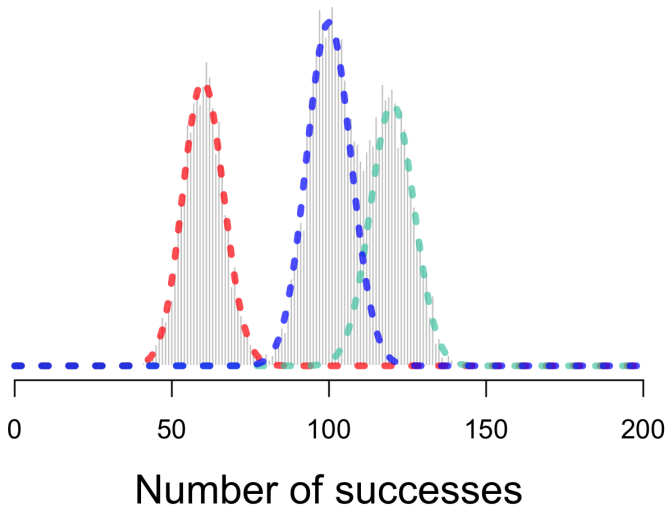
## EM Iteration 7



## EM Iteration 17



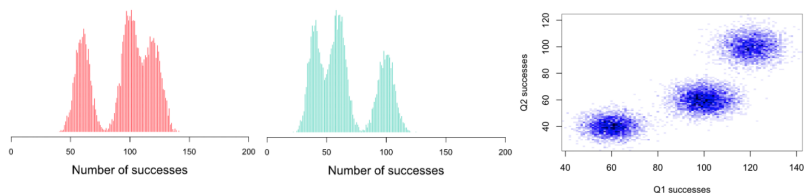
## EM Iteration 37



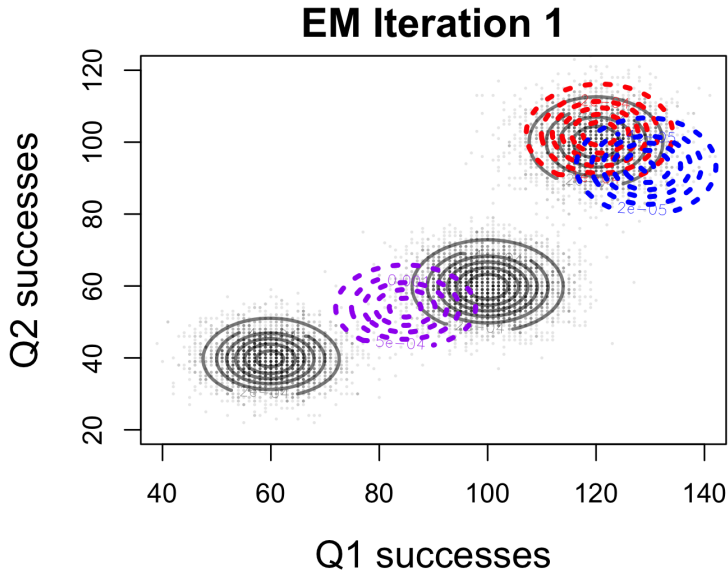
## EM over two questions (novel approach)

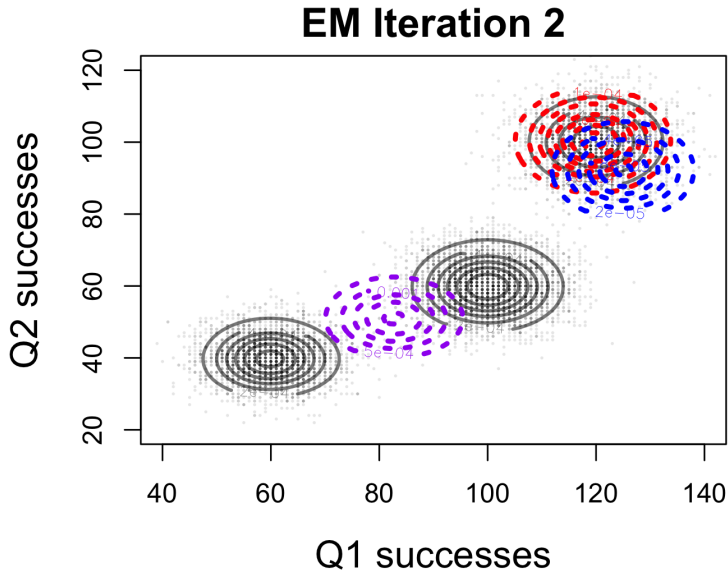
Question 1: I train my students for a professional workplace

Question 2: Learning should always be difficult



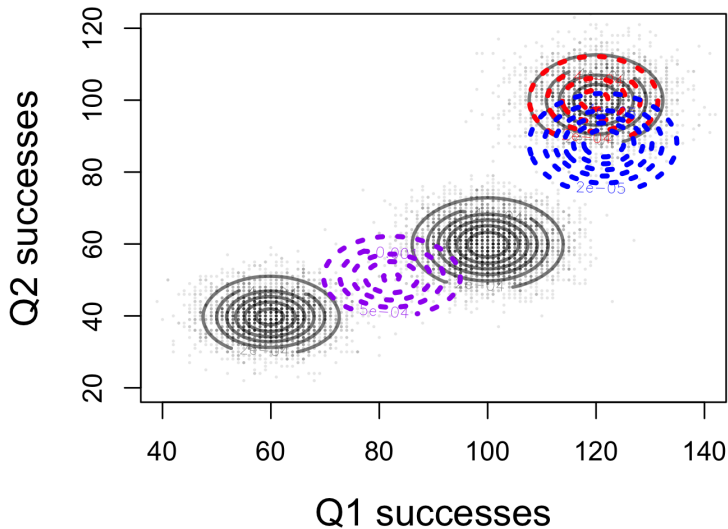
Choose  $K = 3$ . Guess parameters  $\lambda_k$ ,  $\mu_{kq}$ , and  $\nu_{kq}$ .

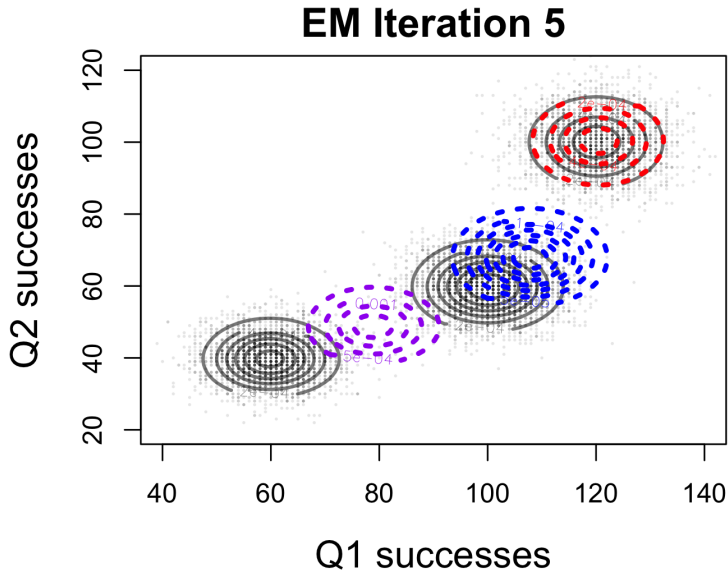


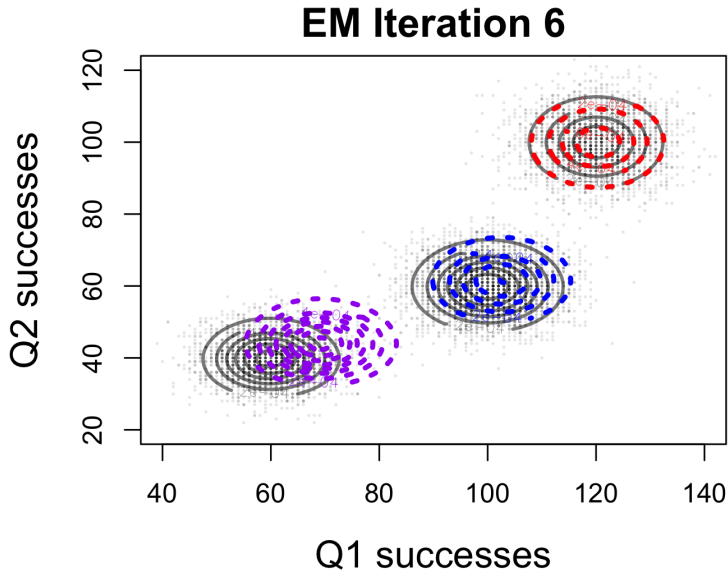




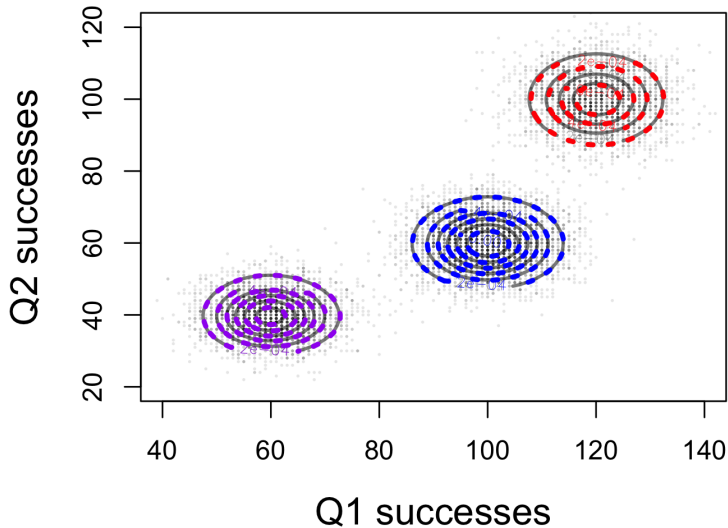
## EM Iteration 3







## EM Iteration 12



When fed  $Q$  questions, the EM converges to local maxima of  $\lambda_k, \mu_{kq}, \nu_{kq}$

1. How many educational ideologies exist among Bowdoin faculty?

1.1  $K$ , to be obtained through a model selection heuristic

2. How are departments mixed by ideology?

2.1 Examine responsibilities  $\gamma_{ik}$ , or  $\hat{z}_i = \arg \max \gamma_{ik}$

3. What parameters define these ideologies?

3.1 Examine learned  $\mu_{kq}, \nu_{kq}$

When fed  $Q$  questions, the EM converges to local maxima of  $\lambda_k, \mu_{kq}, \nu_{kq}$

1. How many educational **political** ideologies exist among Bowdoin faculty **a given electorate**?

1.1  $K$ , to be obtained through a model selection heuristic

2. How are departments **locales** mixed by ideology?

2.1 Examine responsibilities  $\gamma_{ik}$ , or  $\hat{z}_i = \arg \max \gamma_{ik}$

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### Assumptions:

1. Ideologies are latent, consistent sets of principles and beliefs that guide political engagement
  - 1.1 Is the basis for ideology demographic, linguistic, religious, regional, economic?
2. Voting populations are finite mixtures of political cultures, which are informed by ideologies

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Referendum are very direct, not just voting for a political candidate

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### Referendum ballot measure, Maine 2016

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Legalize marijuana for personal use.

3% tax on household income over \$200,000.

Background checks for gun sales and transfers.

Increase minimum wage to \$12 per hour by 2020.

Establish ranked-choice voting.

\$100 million in bonds for transportation projects.

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## Sanity check: Maine case study

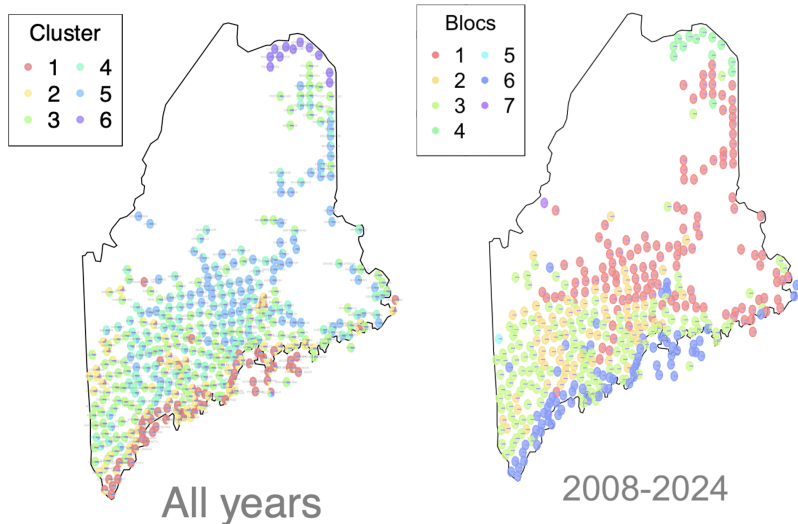


Figure 1: Responsibilities per Maine municipality. Left: Bayesian, 6 weeks of computation. Right: EM, 16.12 seconds of computation.

### **Important exploratory questions:**

1. What is the empirical time complexity of this algorithm?
2. What is the probabilistic accuracy of this inference procedure?

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1. What is the empirical time complexity of this algorithm?
2. What is the probabilistic accuracy of this inference procedure?

## Creating our own data:

1. Draw a single global mixture weight over the  $K$  ideologies
2. Draw a support pattern per latent ideology per question.
3. For each locale, draw a latent ideology label
4. Combine support pattern per ideology with each question, per drawn locale labels

## Spanned parameter sets: <sup>a</sup>

$$M \in \{10, 50, 200, 500, 2000\}$$

$$Q \in \{1, 5, 20, 70\}$$

$$N_{iq} \in \{100, 1000, 5000, 30000\}$$

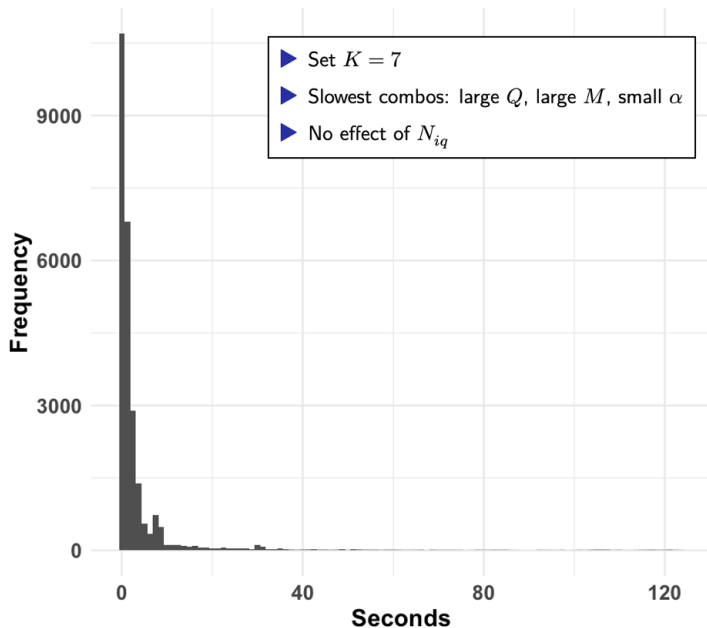
$$\nu_{kq} = 100$$

$$\alpha \in \{0.1, 0.3, 0.5, 0.8\}$$

---

<sup>a</sup>320 unique parameter combos

## Computational time simulations



## Recovering $K$ simulations

1. Fit model over a range of  $K$
2. Compute model selection heuristic BIC
3. Choose  $K$  such that  $BIC(K) - BIC(K - 1) < \epsilon$

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$$Q \in \{1, 5, 20, 70\}$$

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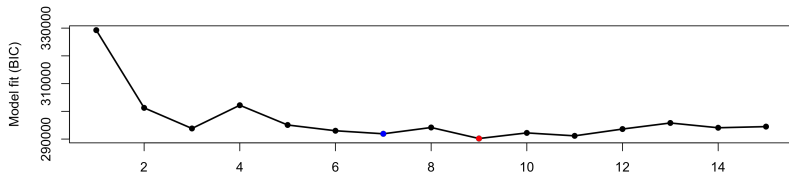
$$\nu_{kq} = 100$$

$$\alpha \in \{0.1, 0.3, 0.5, 0.8\}$$

$$K_{true} \in \{3, \dots, 9\}$$

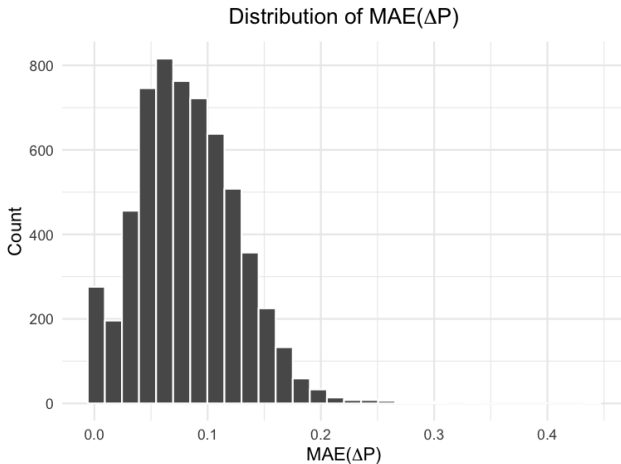
$$K_{search} = 1, \dots, 15$$

$$BIC = \underbrace{\mathcal{K} \ln(M \cdot Q)}_{\text{complexity}} - \underbrace{2 \ln(\mathcal{L})}_{\text{fit}}$$



- ▶ Slowest recovery of 77.5 minutes
  - ▶  $(M, N, Q, \alpha, K) = (2000, 100, 70, 0.5, 3)$
- ▶ Larger  $Q$  improves model ability to find  $K$
- ▶ Model over-predicts  $K$  in most cases
  - ▶ I.e. Jackman on Maine map
- ▶ Note: 1,680,000 EM runs for this study!

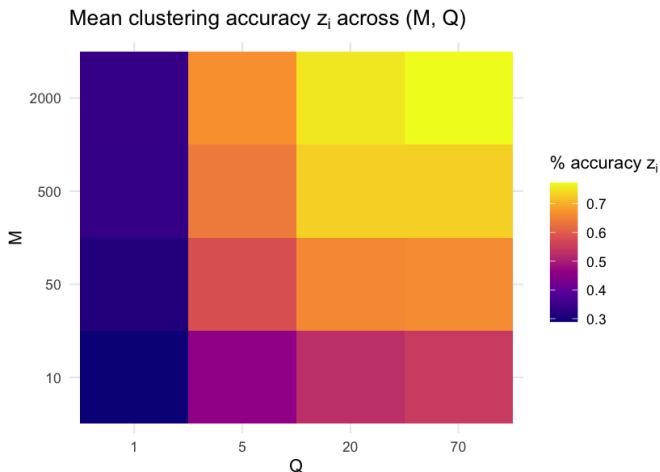
$P_{K \times Q}$  vs  $\widehat{P}_{K \times Q}$ , across usual parameters with  $K \in \{4, \dots, 10\}$



- ▶ On average 8% wrong
- ▶  $Q, M$  have weak effect

## Recovering mixture weights in simulation

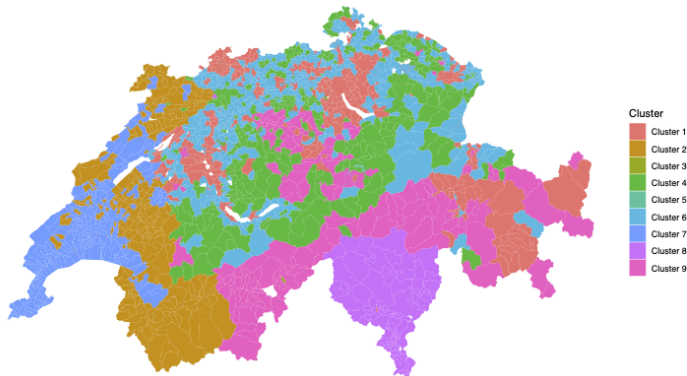
$z_i$  vs  $\widehat{z}_i = \arg \max_k \widehat{\gamma}_{ik}$ , across usual parameters with  $K \in \{4, \dots, 10\}$



► Feature of the algorithm: sparse responsibility vectors



- ▶ Switzerland system of direct democracy
- ▶ Rich longitudinal referendum data
  - ▶ Since 1981, 390 questions across 2048 municipalities (26 cantons)
  - ▶ Bayesian method limits out at  $\sim 500$  municipalities
- ▶ 7.25 minutes to infer model and below hard assignments

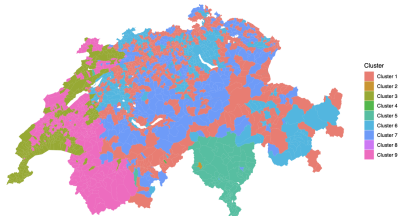


# Consistency across 10-year periods

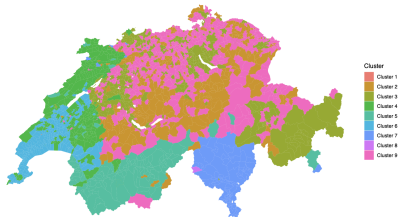
Swiss municipalities: hard cluster assignment (Jan 1981 – Dec 1991)



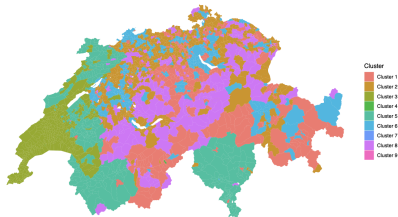
Swiss municipalities: hard cluster assignment (Jan 1992 – Dec 2002)



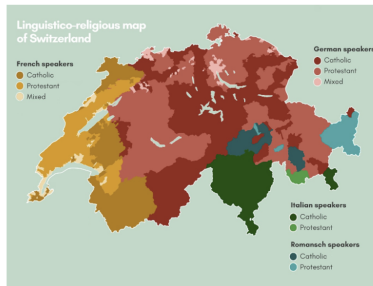
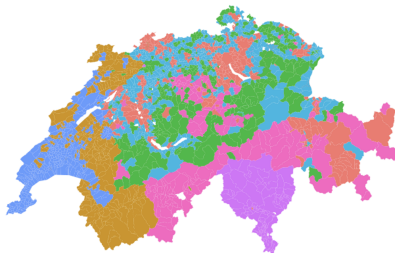
Swiss municipalities: hard cluster assignment (Jan 2003 – Dec 2013)



Swiss municipalities: hard cluster assignment (Jan 2014 – Dec 2026)



## Referendum ideology $\iff$ religious and linguistic patterns



*Map (right) courtesy of CactusMapping*

- ▶ Religion and language are bases for political ideologies!
  - ▶ Derived from settlement patterns?

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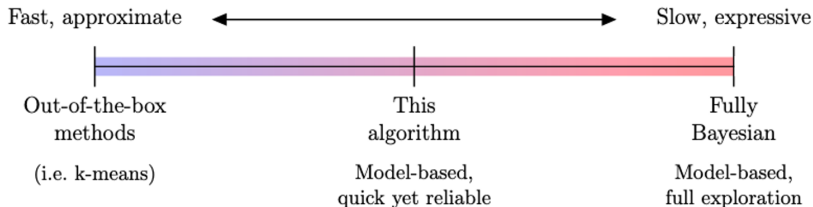
Can we mathematically model voting ideology in referendum? **Yes!**

► Religion and language are bases for political ideologies!

► Derived from settlement patterns?

Can we mathematically model voting ideology in referendum? **Yes!**

Can we pinpoint exactly where ideology comes from? **Not quite...**



A special thanks to Prof. Jack O'Brien and the Mathematics Department



Questions, please!

- ▶ How well does the model handle NA values in data?
  - ▶ Miami referendum data
- ▶ Invariance of method to spatial boundaries?
  - ▶ Canton vs Bezirke vs municipality
- ▶ Methods for more informative exploration of mixture
  - ▶ Laplacian approximation near MLE?